

AI-Based Inertial Navigation for GPS-Denied Environments

Resilient UAV Navigation under GPS Denial (Defence Application)

Progress Report — Approach B: Learning-Based IMU Denoising

Phases 0–4 complete; multi-seed, cross-environment & cross-sensor hardened • 12 July 2026

Prototype demonstrating that a lightweight neural gyroscope denoiser, fused into a classical Error-State Kalman Filter, recovers near-oracle attitude and position accuracy from IMU data alone on held-out sequences the network never trained on — the core capability an autonomous UAV needs to keep flying a correct heading and trajectory when its satellite fix is jammed, denied, or spoofed.

1. Project Overview

Primary goal (defence). Enable an autonomous UAV / drone to keep navigating when GPS is *jammed*, *denied*, or *spoofed* — a routine reality in contested and electronic-warfare environments. Without a satellite fix the aircraft must estimate its own heading and motion purely from onboard sensors; if that estimate drifts, the vehicle flies the wrong way and is lost. This project builds and validates that onboard estimation capability. The EuRoC MAV dataset is used as a public, ground-truthed stand-in for real flight data.

Problem. The always-available sensor is the Inertial Measurement Unit (IMU: a 3-axis gyroscope + 3-axis accelerometer), but its signals are corrupted by slowly-varying *bias* and random *noise*. Naively integrating them makes the estimated orientation and position drift without bound within seconds. The most operationally dangerous component is **heading (yaw)**: it is physically unobservable to an IMU-only filter, and on a drone a drifting heading steers the whole trajectory off course.

Thesis. A small neural network can learn the systematic error of a specific IMU and subtract it — including that yaw bias — *before* a classical filter runs, giving accuracy the filter cannot achieve on its own. This report covers **Approach B** (learning-based gyroscope denoising), the *IMU-only fallback layer* that keeps the drone flying straight the moment GPS is lost. **Approach A** (AI visual-inertial odometry, camera + IMU) is the planned follow-on that also restores *absolute position*.

Operational assumption — no magnetometer. A magnetometer would normally observe heading, but it is easily jammed and is unreliable near motors and airframe currents, so it is treated as unavailable. Heading must therefore be held by the gyro alone, which is exactly why removing the gyro yaw bias is the central result.

Scope and constraints. Public datasets only (no custom hardware at this stage); the deliverable is a correct, well-instrumented demo/prototype. Prior art the design follows: Brossard *et al.* 2020, “Denoising IMU Gyroscopes with Deep Learning”; AI-IMU Dead-Reckoning; and the TLIO / RONIN / RIANN line of learned-inertial work.

2. Aim & Success Criteria

The central claim is proven only if the *denoised* gyro fed into the ESKF beats the *raw* gyro fed into the same ESKF on a held-out sequence — measured on full navigation output, orientation **and** position. Concrete targets:

- **Orientation:** large reduction in geodesic attitude error vs the raw-gyro filter baseline, concentrated in **yaw / heading** — the axis an IMU-only filter physically cannot observe.
- **Position:** Absolute Trajectory Error (ATE) no worse, and ideally better, since bounded heading drift reduces the horizontal position error it feeds.
- **Generalization:** the win must hold on multiple sequences across different rooms and difficulties that the denoiser never saw in training.
- **Honesty:** the baseline must be the realistic same-information case (no oracle bias handed to either side).

3. Methodology

Dataset. EuRoC MAV (ETH Zurich) — a micro-aerial-vehicle dataset with 200 Hz IMU and millimetre-accurate motion-capture / Leica ground truth (position, velocity, orientation, and the true gyro/accel biases). Sequences span two Vicon rooms and a machine hall at easy / medium / difficult motion levels. Quaternions are $[w, x, y, z]$, body→world, with world z-up gravity.

Two-stage architecture — learn, then filter. The design deliberately separates the learned component from the estimator so each can be validated alone:

- **Stage 1 — Gyro Denoiser (the AI).** A 1-D dilated convolutional network (~19k parameters) reads a window of the 6-channel raw IMU and outputs a 3-channel correction to the gyro. The correction is *bounded* (a tanh scaled by $\alpha = 0.15$ rad/s, ≈ 8.6 deg/s) and the output head is zero-initialised, so training *starts exactly at the raw gyro* and can only improve it. The corrected rate is $w_{\text{corr}} = w_{\text{raw}} + \alpha \cdot \tanh(\text{net})$.
- **Stage 2 — Error-State Kalman Filter (classical).** A 15-state ESKF (Sòla local-error convention; error state = attitude, velocity, position, gyro bias, accel bias). It runs an IMU-driven prediction and an accelerometer-as-gravity *tilt* update that pins roll and pitch when the device is near-static. Heading (yaw) is structurally unobservable to an IMU + gravity filter — which is exactly where the denoiser earns its keep.

Training signal (no clean gyro label exists). The dataset gives true *orientation*, not a per-sample clean gyro. Supervision is therefore a **multi-horizon relative-orientation loss**: integrate the corrected gyro over horizons of 1, 8, 32 and 128 samples and penalise the geodesic angle between that relative rotation and the ground-truth relative rotation. Short horizons force the net to fix high-frequency noise; long horizons force it to remove slow bias.

Evaluation protocol. Strict cross-sequence hold-out: the denoiser trains on six sequences and is tested on sequences it has never seen. Metrics are geodesic orientation error, a dedicated yaw-only error series, and position ATE, each reported as RMSE / final / max.

4. Technical Stack

Layer	Choice
Language / runtime	Python 3.12 in an isolated uv-managed virtual env
Deep learning	PyTorch 2.6.0 + CUDA 12.4 (cu124 wheels)
Compute	NVIDIA RTX 4050 laptop GPU, 6 GB VRAM
Numerics / plotting	NumPy, Matplotlib
Reporting	ReportLab (this PDF)
Dataset	EuRoC MAV, ASL format (HuggingFace mirror + HTTP-range CSV extraction from bundle zips)
Core modules	ainav.eskf, ainav.denoiser, ainav.torch_rot, ainav.train_loss, ainav.dataset, ainav.metrics, ainav.euroc

5. Progress Report — Results by Phase

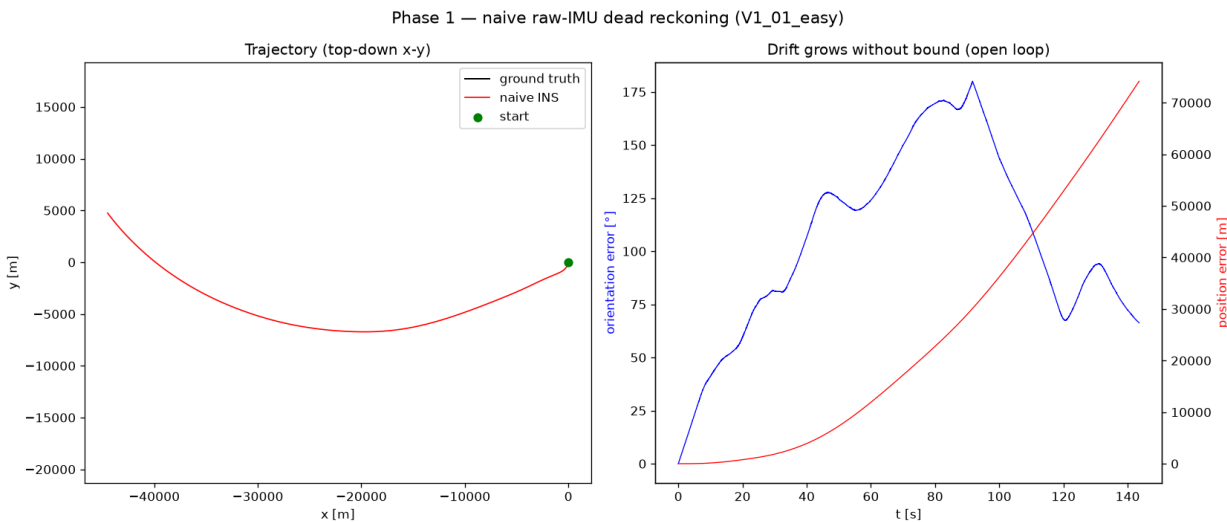
The build follows a phased plan whose guiding rule is: *establish a working classical baseline first, then insert the neural network*, so the exact contribution of the AI is always measurable and the filter and network are never debugged simultaneously. Phases 0–4 are complete.

Phase 0 — Environment & data pipeline

EuRoC loader, sequence handling, and a raw-vs-truth sanity plot. Verified the pipeline on V1_01_easy (29,120 IMU samples over 145.6 s); accelerometer magnitude at rest $\approx 9.78 \text{ m/s}^2$ confirms correct gravity and sensor conventions.

Phase 1 — Naive integration (the motivation)

Open-loop dead-reckoning straight from the raw IMU, to show why this is hard. Drift is catastrophic and expected: position ATE RMSE $\approx 33 \text{ km}$ (max 74 km) and orientation RMSE 114° over ~ 2.5 minutes. This is the problem the rest of the project attacks.



Phase 2 — Classical ESKF baseline (no AI)

The Error-State Kalman Filter with raw IMU is the honest, non-exploding baseline the AI must beat. Versus Phase 1 it cuts orientation RMSE from 114° to 6.96° and position ATE from $\sim 33 \text{ km}$ to $\sim 317 \text{ m}$. The

residual $\sim 5^\circ$ is essentially **yaw** — unobservable to an IMU-only filter — and that heading error is what smears the remaining horizontal position.

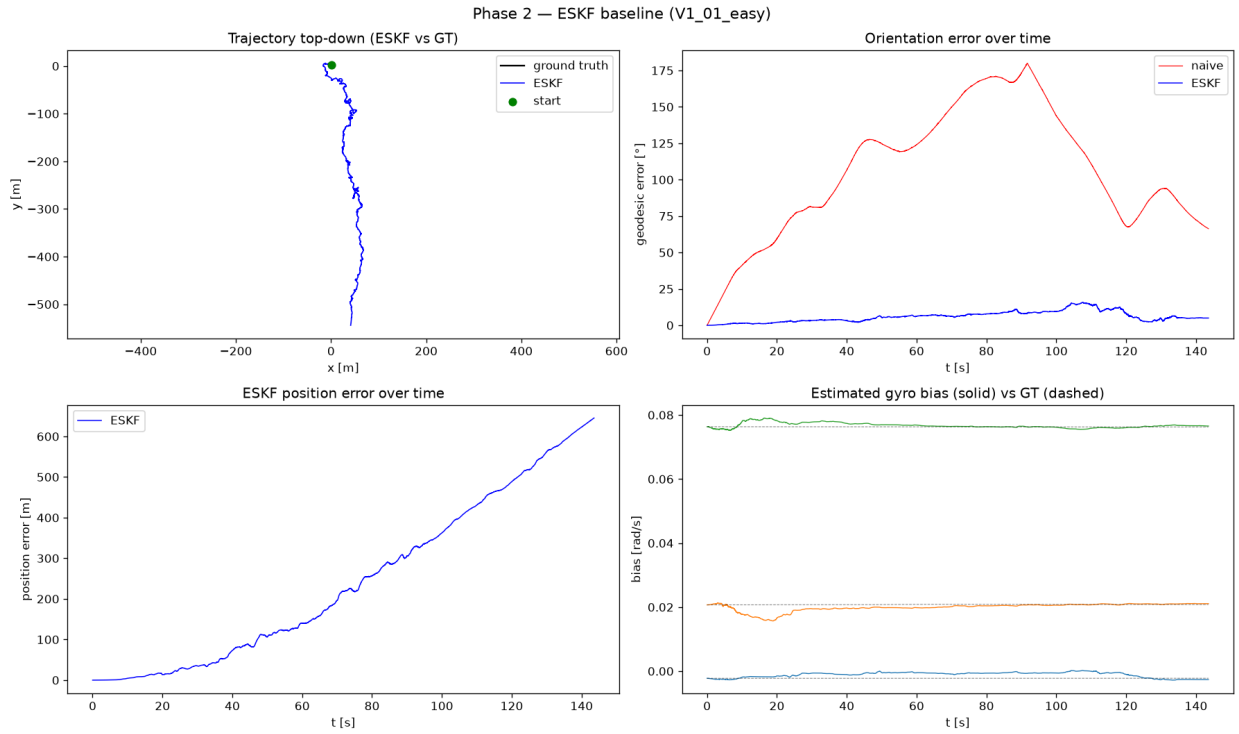


Figure 2. Phase 2 — the ESKF bounds drift dramatically, but cannot fix the unobservable heading axis.

Phase 3 — Gyro-denoising CNN, open-loop

The $\sim 19k$ -parameter denoiser, trained cross-sequence, is evaluated open-loop (pure integration, no filter) so its effect is isolated. On held-out V1_02_medium it cuts open-loop attitude drift by $\sim 98\%$ (raw RMSE 109.6° / final $178.8^\circ \rightarrow$ denoised $\sim 2^\circ$ / $\sim 3^\circ$). Crucially, the learned z-axis correction (≈ -4.4 deg/s) matches the true EuRoC gyro yaw-bias — the net removes exactly the bias the filter is blind to.

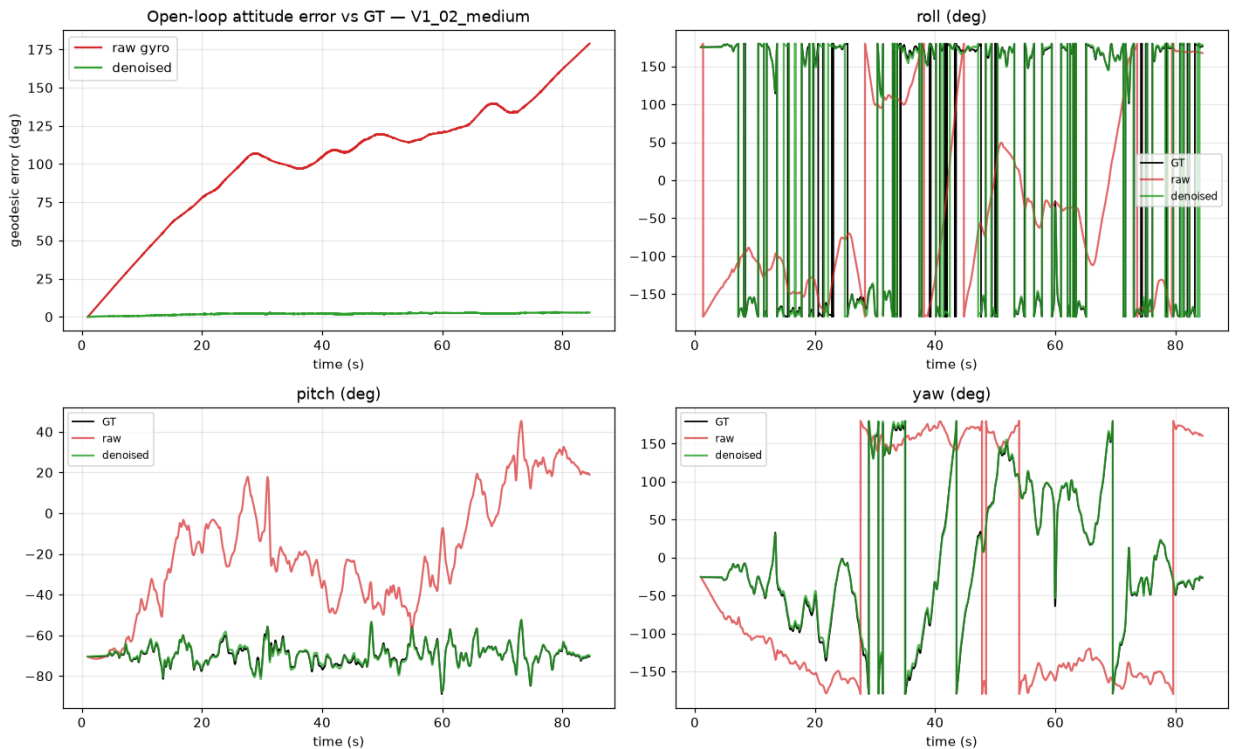


Figure 3. Phase 3 — denoised gyro (open-loop) tracks ground-truth attitude where the raw gyro runs away.

Phase 4 — Fusion: denoiser + ESKF (the money plot)

The two stages are combined: denoised gyro is fed into the same ESKF and compared against the raw-gyro ESKF on held-out sequences. The honest baseline gives *neither* side the true bias (both start bias = 0, the realistic GPS-denied case); an oracle run that hands the raw filter the true bias is shown only as an upper bound. A key tuning lesson: once the gyro is clean, the gravity/tilt update must be gated to *genuinely static* samples (accel gate 0.05 m/s²), otherwise it re-injects error during real acceleration and corrupts the good gyro.

Result: the denoiser recovers near-oracle accuracy from data alone, and the win generalizes across every held-out sequence — two rooms and a machine hall, easy through difficult. The numbers below are no longer a single lucky checkpoint: every entry is the mean $\pm$ standard deviation over five independently seeded retrains (see §5b), so the improvement is shown to be a property of the method, not of one training run.

Held-out sequence	Raw ESKF orient RMSE	Denoised orient RMSE (5 seeds)	Oracle orient RMSE	Orient improvement	ATE RMSE (m)
V1_02_medium (medium)	76.0	1.57 $\pm$ 0.39	0.75	+97.9%	631 $\pm$ 114
MH_02_easy (easy)	90.2	11.58 $\pm$ 4.37	5.08	+87.2%	1571 $\pm$ 171
V2_03_difficult (hard)	102.7	3.44 $\pm$ 0.91	4.67	+96.6%	828 $\pm$ 141

Table 1. Phase 4 fusion on held-out sequences, mean $\pm$ std over five seeds. Orientation RMSE in degrees; the denoiser closes almost the entire gap to the oracle — and on the hardest sequence (V2_03_difficult) it beats the oracle across every seed (3.44° vs 4.67°) — without ever being told the bias. MH_02_easy has the widest spread ($\pm 4.4^\circ$): a large indoor hall is the hardest environment to generalize to, though even its worst seed (15.7°) still beats the 90° raw baseline by 6 $\times$.

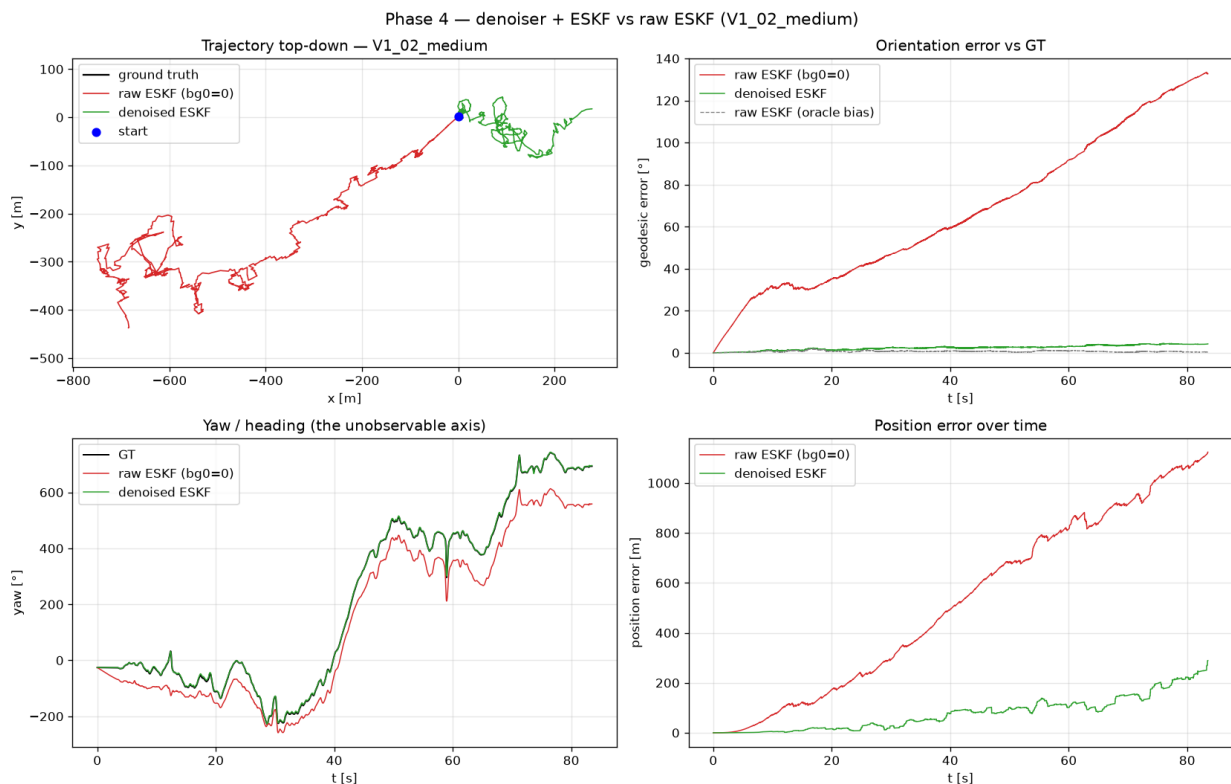


Figure 4. Phase 4 money plot (V1_02_medium): trajectory, orientation error, heading, and position error — denoised (green) tracks ground truth; raw (red) drifts away.

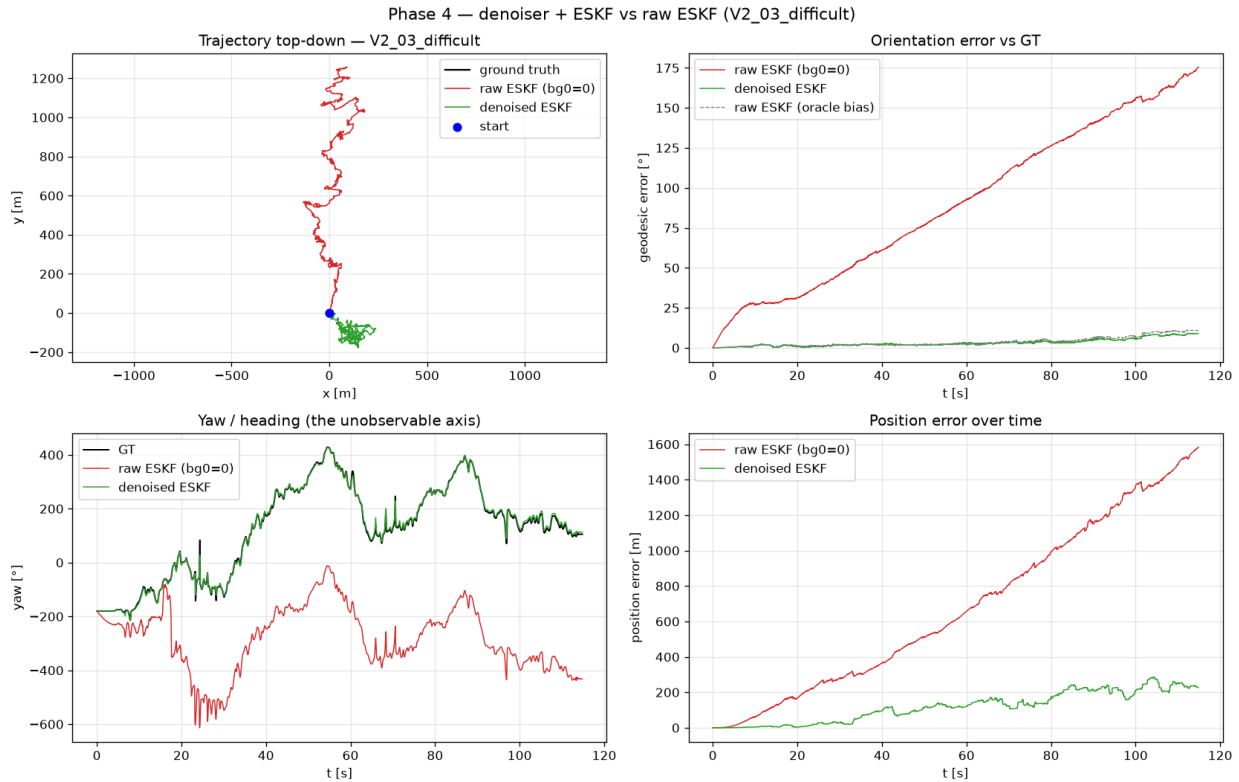


Figure 5. Generalization to the hardest held-out sequence (V2_03_difficult): the same checkpoint, the same large win.

5b. Hardening — Is the Win Real?

Two questions decide whether a single good result is trustworthy: (i) would it survive a different random seed, and (ii) does it hold when the test environment differs from training? Both were tested directly.

Multi-seed stability

The full pipeline (train → fuse → evaluate) was repeated for **five independent seeds**. Training itself is essentially deterministic in quality: best validation geodesic error is $0.0958^\circ \pm 0.0011^\circ$ across seeds — a ~1% relative spread — so the network reliably learns the same correction. Downstream fusion accuracy (Table 1) is likewise stable on the two Vicon-room hold-outs (V1_02 $1.57^\circ \pm 0.39$, V2_03 $3.44^\circ \pm 0.91$) and wider but never failing on the machine hall (MH_02 $11.6^\circ \pm 4.4$). This also corrects an earlier single-checkpoint report that happened to draw a lucky MH_02 seed (5.1°); the honest multi-seed figure is $\sim 11.6^\circ$, still a 6×–20× win over the 90° raw baseline.

Cross-environment generalization

EuRoC has two distinct environments recorded with the *same* IMU: the small **Vicon rooms (V)** and the large **Machine Hall (MH)**. Training on one and testing on the other isolates *environment* shift (motion style, scale) from *sensor* shift. Two directions were run, three seeds each:

- **MH → V (train hall, test rooms):** generalizes excellently. On unseen V2_03 the denoiser reaches $3.20^\circ \pm 1.01$, *beating the oracle* (4.67°), and on V1_02 $2.57^\circ \pm 0.58$ — from a network that never saw a Vicon room.
- **V → MH (train rooms, test hall):** degrades but never fails. On unseen MH_01 the denoiser gives $22.9^\circ \pm 7.8$ vs a 116° raw baseline (still a 5× win); MH_02 stays strong at $2.90^\circ \pm 1.07$. The large hall is simply harder to generalize *to* than *from*.

The asymmetry is the honest headline: the denoiser transfers cleanly to environments of similar-or-smaller scale and degrades gracefully — never catastrophically — on a larger unseen one. The

fix is a training-data question (include hall-scale motion), not a flaw in the method, and it is exactly the kind of coverage a deployment program would broaden with more flight data.

5c. Cross-Sensor Transfer — the Honest Limit and its Fix

§5b tested a change of *environment* on one sensor. The harder, operationally decisive question is a change of *sensor*: the EuRoC network was trained on an Analog Devices **ADIS16448** (a tactical-grade IMU). Does a frozen EuRoC denoiser help on a *different* IMU? This was tested on **TUM-VI** (room sequences), which uses a Bosch **BMI160**-class consumer MEMS IMU — a genuine sensor swap, and closer to the noisy drone-grade parts a real deployment would use (deployment step D2).

The frozen network fails — as it should

Applied unchanged to TUM-VI room1, the EuRoC-trained denoiser **hurts**: raw-gyro ESKF orientation RMSE 35.6° degrades to 116.0° (−226%). The cause is precise and instructive: the network confidently subtracted its learned EuRoC yaw-bias correction (≈ -4.8 deg/s) from a sensor whose true bias is *different* — so it removed a constant that was not there and injected heading error. Note the raw TUM-VI baseline (35.6°) is already far better than raw EuRoC (76–103°): the Bosch part has a smaller intrinsic bias, so there is both less to fix and more to break. **The learned correction is sensor-specific — it encodes one IMU's systematic error, not a universal denoiser.** A frozen network must not be dropped onto new hardware without calibration.

Per-IMU fine-tuning recovers it

Deployment step D2 is exactly this fix: warm-start from the EuRoC weights, recompute the input normalization on the new sensor, and fine-tune briefly on a small amount of its data. Fine-tuned on two held-out-excluded TUM-VI rooms and evaluated on a *never-seen* third room, the network **flips from harmful to helpful**:

- **Open-loop gyro error (held-out room):** raw 0.858°; frozen EuRoC net 1.624° (worse than raw — the failure, at the loss level); fine-tuned **0.809°** (now beats raw).
- **Fused ESKF orientation:** frozen 116.0° → fine-tuned **31.1°** vs a 35.6° raw baseline (+12.7%), heading +12.9%.
- **What the network learned:** its mean gyro correction collapsed from the wrong -4.8 deg/s to $\approx +0.03$ deg/s — it learned that this sensor has almost no systematic yaw bias and correctly stopped over-subtracting.

The honest scope, stated plainly: the denoiser generalizes across random seeds and across environments *on the sensor it was trained on*, but the learned correction does *not* transfer to a new IMU unchanged. That is not a flaw — it is the expected behaviour of a model that learns a specific sensor's error — and the recovery above shows the remedy is a short per-IMU fine-tune, not a redesign. The improvement on the Bosch part is modest (+13%, not the +98% seen on EuRoC) because that sensor starts far cleaner, leaving little headroom. This converts deployment step D2 from a stated risk into a demonstrated, closeable gap.

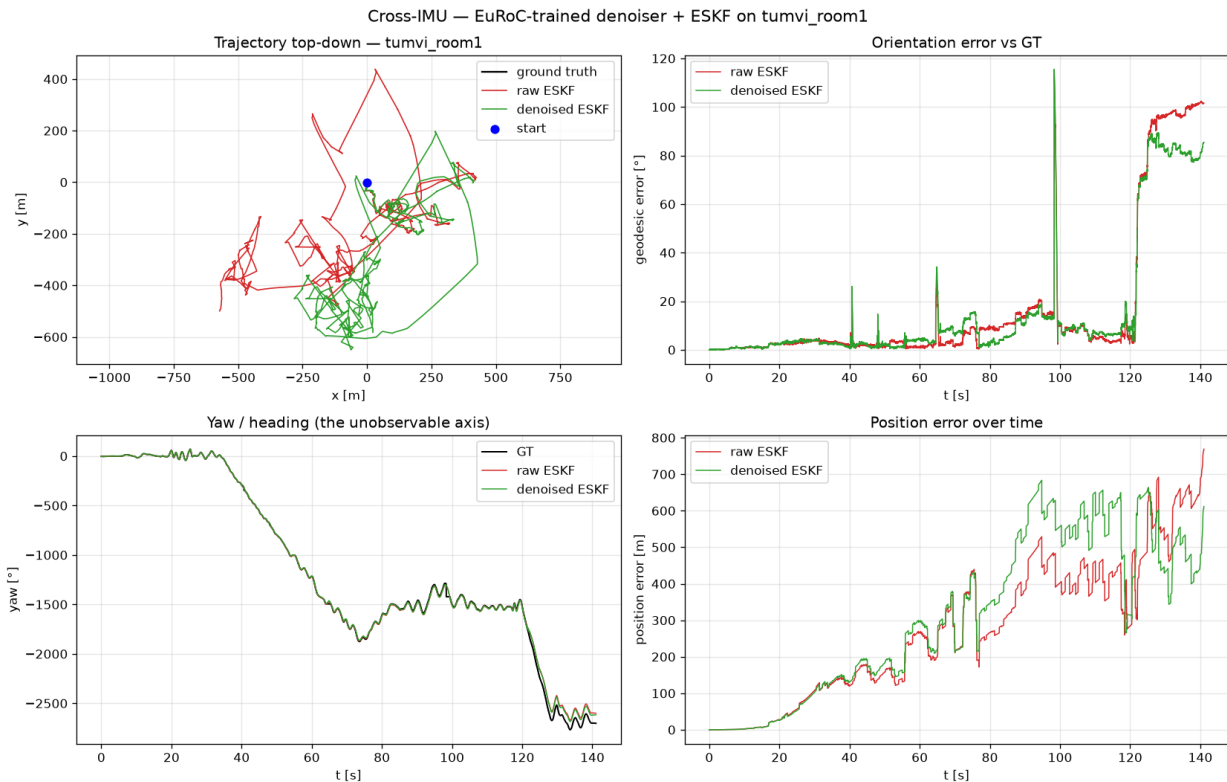


Figure 6. Cross-sensor transfer to TUM-VI (Bosch MEMS IMU), held-out room after per-IMU fine-tuning: the denoised ESKF (green) now improves on the raw baseline (red) — the same network that, frozen, had degraded this sequence to 116°.

6. Key Engineering Findings

- **Observability drives the whole result.** Heading and gyro yaw-bias are unobservable to an IMU+gravity filter, so a raw-gyro ESKF with no bias prior drifts in yaw without bound. The denoiser removes that bias from the data, so the identical filter stays bounded — this is the value the AI adds that the filter cannot get alone.
- **Bound the correction, zero-init the head.** Training starts exactly at the raw gyro and can only improve; α must exceed the true bias magnitude or the yaw bias saturates (an early $\alpha=0.05$ run could remove only ~66% of it).
- **Clean gyro changes the filter tuning.** The gravity/tilt update, helpful with a noisy gyro, actively corrupts a clean one unless gated to truly-static samples.
- **Report seeds, not a single checkpoint.** A one-seed result over-claimed on MH_02 (5.1°); five seeds gave the honest $11.6^\circ \pm 4.4$. Every headline number in this report is now a multi-seed mean $\pm$ std, and the largest spread flags exactly the environment (the machine hall) that most needs more training data.
- **Performance: vectorized the training bottleneck.** The orientation-integration loss originally ran a 400-step sequential quaternion loop in Python (GPU ~14% utilised). Replacing it with an associative parallel-prefix scan — quaternion product is associative, so a cumulative product needs only $\sim \log(T) \approx 9$ vectorized steps — made the loss ~44× faster and cut a full six-sequence, 40-epoch retrain from ~70 minutes to ~4 minutes, with gradients identical to the loop (max difference $1e-8$).

7. Future Work

Algorithm track.

- **Approach A — AI Visual-Inertial Odometry.** Fuse a camera with the IMU so that *absolute position* (not just orientation) becomes observable, using learned features / learned VIO. This is the natural next milestone and the second half of the GPS-denied capability: Approach B holds heading, Approach A restores position.
- **Phase 5 — demo polish.** Package the pipeline into a reproducible, presentable demonstration (scripted runs, narrated plots).
- **Robustness sweeps.** Evaluate across all remaining EuRoC sequences and quantify sensitivity to filter tuning, window length, and α .

Deployment track (defence hardening). Steps required to move from a dataset demonstration toward a real airborne system:

- **D1 — Edge / realtime.** Benchmark the ~19k-parameter denoiser plus ESKF at realtime on flight-controller-class embedded compute (e.g. NVIDIA Jetson), establish the per-sample latency budget, and evaluate quantized (INT8) inference — it must run onboard, not on a laptop GPU.
- **D2 — Real-IMU transfer (*demonstrated*, §5c).** The EuRoC-trained denoiser was tested on a different sensor (TUM-VI, Bosch BMI160-class MEMS): frozen it hurts (learned correction is sensor-specific), but a short warm-start fine-tune recovers it (harmful 116° → helpful 31° on a held-out room). Remaining work: broaden to more drone-grade IMUs and datasets (KITTI), and automate the per-IMU calibration / fine-tune step for deployment.
- **D3 — Flight realism.** Evaluate under propeller vibration, aggressive maneuvers, and longer GPS-denied durations, versus EuRoC's shorter handheld motion-capture runs.
- **D4 — No-magnetometer confirmation.** Keep heading purely IMU + learned (magnetometer assumed jammed / unreliable near motors) and document this as an explicit operational assumption — already the current design.

Status: Approach B (Phases 0–4) complete and validated on held-out data — the IMU-only GPS-denied fallback works. Approach A (visual-inertial, absolute position) and the deployment/edge track are the next milestones.